SALESFORCE ADMIN COURSE • EPISODE 35

Object-Level Security & CRUD Permissions

Mastering Salesforce Data Authorization: CRUD Operations, View All / Modify All Superpowers, System Permissions vs. Object Permissions, Tab Visibility, and Hands-on Testing with Custom Objects.

- 🔑 Core Question: "What can you do with this object?"
- 📝 CRUD: Create, Read, Edit, and Delete
- ⚡ View All & Modify All Object Permissions
- 🌐 System Superpowers: View All Data vs Modify All Data
- 👁️ Tab Visibility vs Object Access Truths
- 🧪 Hands-On Testing Lab with Employee__c

Created by Sachin • Salesforce Certified Administrator Series • Duration: ~20–25 Mins

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1. Introduction: Object-Level Security in the Hierarchy

In the previous episode, we learned how **Profiles**, **Permission Sets**, and **Permission Set Groups** structure permissions across users.

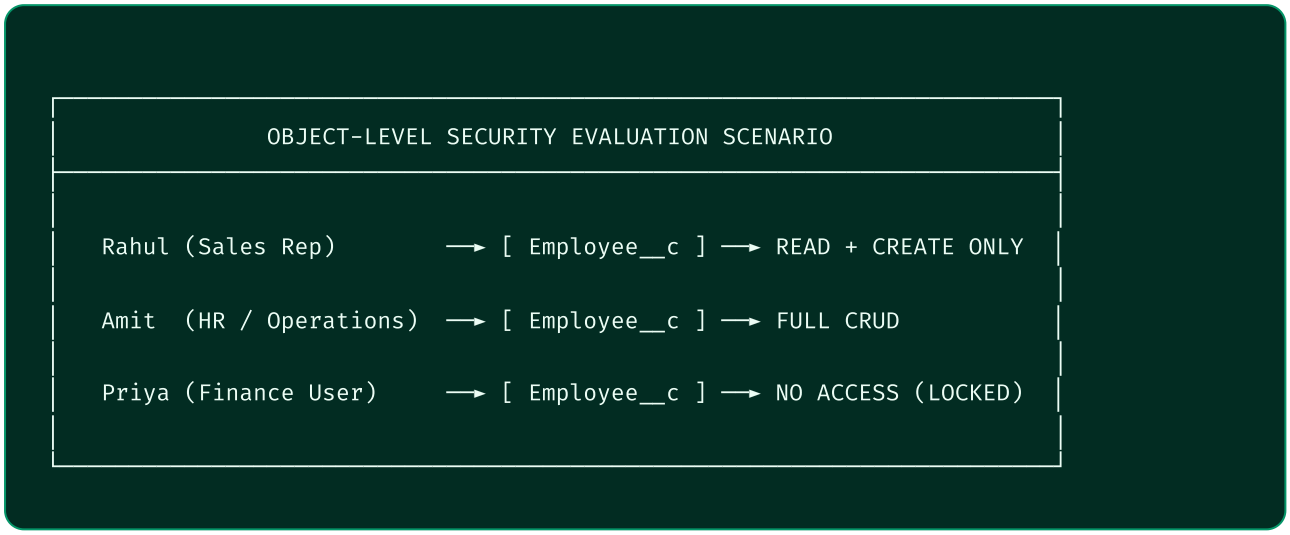
Now we take a massive step forward into the actual data security model. While organization-level security answers *"Can this user enter Salesforce?"*, **Object-Level Security** answers the second fundamental question:

> *"What can this user actually DO with an entire table/object of data?"*

Even if multiple users have active Salesforce accounts and log in through the same portal, their ability to interact with specific objects (like Leads, Accounts, Opportunities, or custom objects like `Employee__c`) can be completely different.

2. What is Object-Level Security?

Object-Level Security acts as the first internal gatekeeper to your data tables. It controls whether a user can access a particular standard or custom object at all, and what baseline operations they can execute on its records.



If a user has **No Access** to an object:

- They cannot see the object in standard search or lookup fields.
- They cannot run reports on that object.
- They cannot access record URLs directly (they will receive an "Insufficient Privileges" error).
- The object does not appear in their navigation or App Launcher.

3. CRUD Permissions Explained

The standard foundation of object permissions is universally referred to as **CRUD**:

C — Create

Allows users to make **new records** of this object (displays the "New" button on list views and lookups).

R — Read

Allows users to **view records** of this object. Without Read, no other CRUD permission is possible!

U — Update (Edit)

Allows users to **modify existing records** of this object (displays the "Edit" button and inline editing pencil).

D — Delete

Allows users to **permanently remove** records of this object (displays the "Delete" action button).

> **Dependency Rule:**
> You cannot grant **Edit** or **Delete** without granting **Read** access. Salesforce will automatically enforce this dependency in the UI when configuring profiles and permission sets.

4. Object Superpowers: View All & Modify All

Beyond standard CRUD, Salesforce provides two advanced object-level permissions that completely override record-level sharing for that specific object:

View All (Object-Level)

Grants the user read-only access to **EVERY single record** of that object across the entire Salesforce org, *regardless of who owns the record, Organization-Wide Defaults (OWD), Role Hierarchy, or Sharing Rules*!

Modify All (Object-Level)

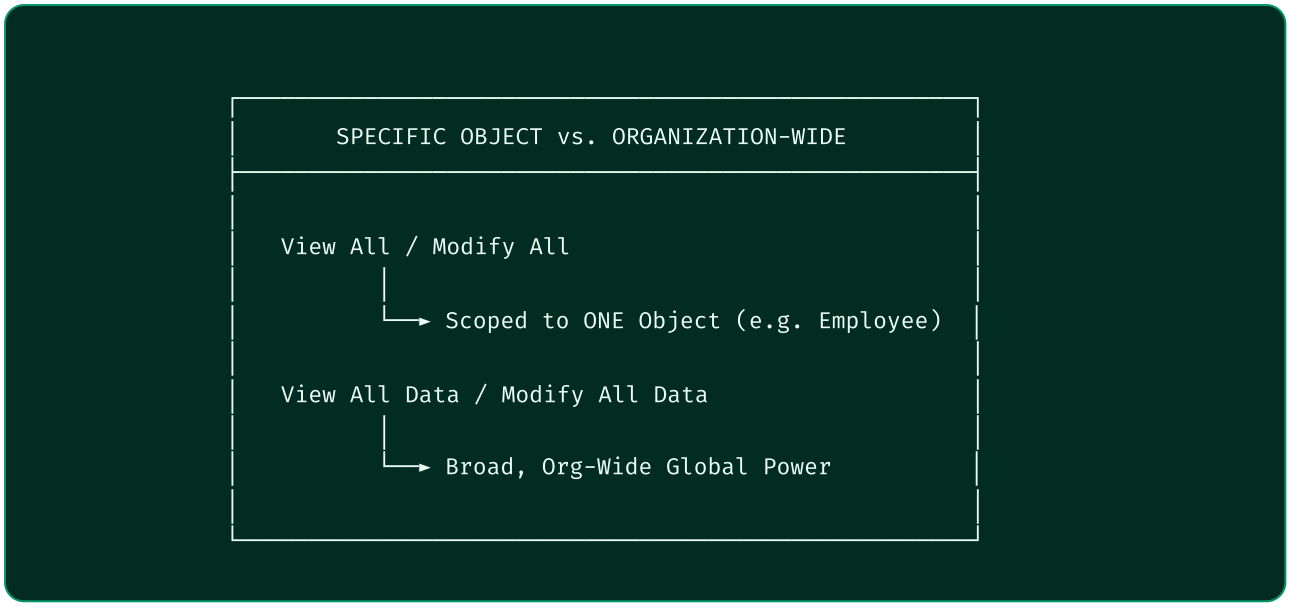
Grants **full CRUD control over EVERY record** of that object across the entire org (view, edit, delete, transfer ownership, and approve), completely bypassing all sharing and ownership barriers!

High Privilege Warning:
View All and **Modify All** should be granted with extreme caution! They are typically reserved for business unit administrators, compliance auditors, or automated integration users.

5. Object vs. System Permissions: The Critical Exam Distinction

One of the most frequently failed concepts on the Salesforce Certified Administrator exam is confusing **View All / Modify All** with **View All Data / Modify All Data**:

PERMISSION NAME	TYPE	SCOPE OF ACCESS	IMPACT & USE CASE
View All	Object Permission	Single specific object (e.g. `Opportunity`)	Allows reading all records of *that object only*, bypassing sharing rules.
Modify All	Object Permission	Single specific object (e.g. `Opportunity`)	Allows viewing, editing, deleting all records of *that object only*.
View All Data (VAD)	System Permission	**ALL Objects across entire Org**	Global read access across every standard and custom object in Salesforce.
Modify All Data (MAD)	System Permission	**ALL Objects across entire Org**	Global god-mode administrative power across all objects. Modifies any record anywhere.



6. Where are Object Permissions Configured?

Just as we established in Episode 34, object permissions are managed through the three core authorization mechanisms:

1. PROFILE

Sets the mandatory baseline CRUD for the user's role.

➕ **combined with**

2. PERMISSION SET

Additively grants extra CRUD rights (e.g. adding Delete to a user whose profile has no Delete).

📦 **bundled into**

3. PERMISSION SET GROUP

Delivers combined object permissions to specific job personas.

Effective Object Access Formula:

`Effective Object Access = Profile CRUD ∪ Permission Set CRUD ∪ PSG CRUD`

Salesforce grants the union of all allowed permissions across all active assignments.

7. Tab Visibility vs. Object Permissions

Beginners frequently assume that hiding a tab removes access to an object, or that adding a tab to a navigation bar gives users CRUD rights. **This is completely false!**

📑 Tab Visibility

Controls whether the object's tab appears on the navigation bar (*Default On*, *Default Off*, *Tab Hidden*). It is purely a **Navigation & UI preference**.

🛡️ Object Permissions (CRUD)

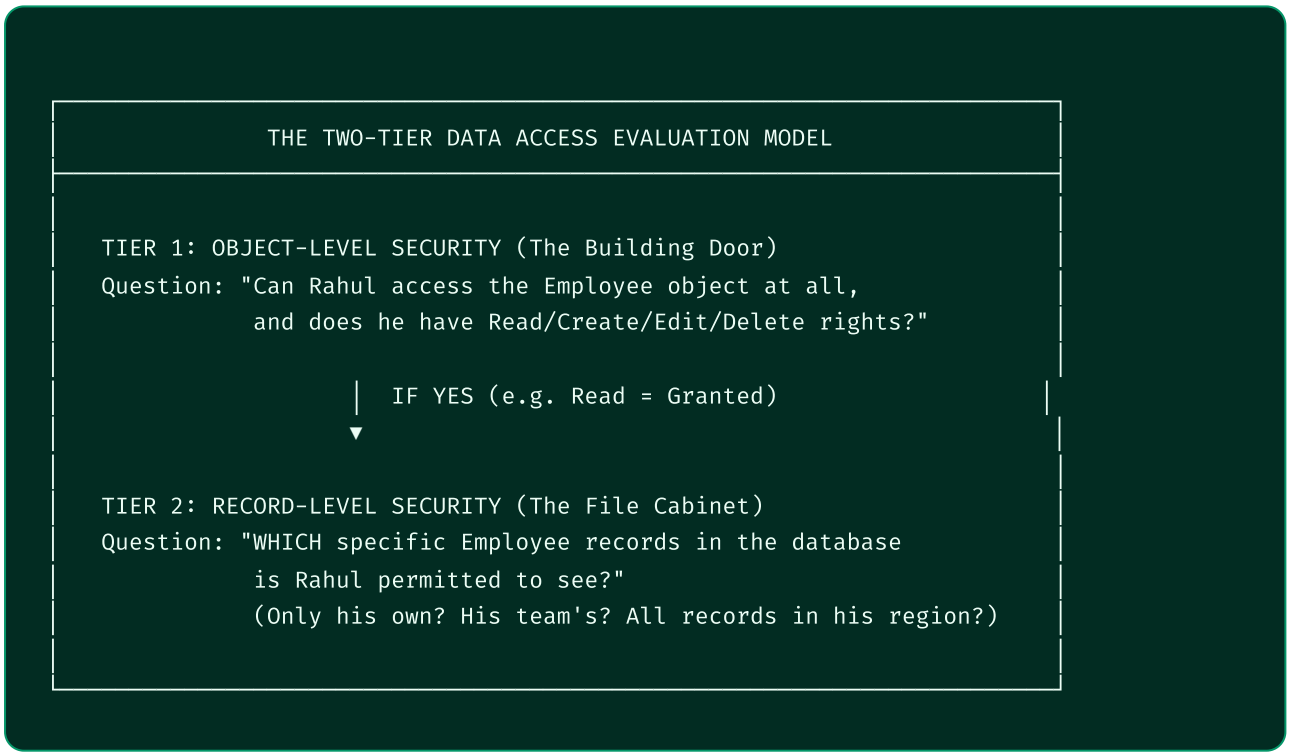
Controls whether the database allows the user to query, view, insert, update, or delete records. It is a **Security & Authorization control**.

Key Truth: Tab Visibility ≠ Object Permissions

- If Tab is **Hidden**, but the user has **Read** permission, they can still view records via Global Search, Related Lists, Reports, or direct URLs.
- If Tab is **Default On**, but the user has **No Object Access**, clicking the tab results in an empty, inaccessible view or permission error.

8. Object-Level Access vs. Record-Level Access

To master Salesforce security, you must always maintain this razor-sharp distinction between the **Object Gate** and the **Record Room**:



If the Object door is locked (No Read permission), Record-Level sharing rules don't even matter—the user cannot see any records regardless of sharing!

9. Hands-On Practical Lab: Building & Testing Employee__c

Follow this step-by-step practical implementation to build a custom object and verify object security across three distinct user perspectives:

STEP 01 — **Create Custom Object: Employee__c**

```
Setup ➔ Object Manager ➔ Create ➔ Custom Object
```

Enter the object definition details:

- **Label:** `Employee` | **Plural Label:** `Employees`
- **Object Name:** `Employee` (API: `Employee__c`)
- **Record Name:** `Employee Name` (Data Type: *Text*)
- Check: *Allow Reports*, *Allow Activities*, *Track Field History*.
- Check: **Launch New Custom Tab Wizard after saving this custom object**.

Select a Tab Style (e.g., *People*) and save the tab.

Add Custom Fields under Fields & Relationships:

- `Department` (Picklist or Text)
- `Designation` (Text)
- `Salary` (Currency)
- `Email` (Email)

STEP 02 — **Configure Rahul's Baseline Access (Read + Create Only)**

```
Setup ➔ Profiles ➔ [Rahul's Profile / Sales User Profile] ➔ Object Settings ➔ Employees
```

Click **Edit** and configure the permissions for `Employee`:

- Read: Checked ✅
- Create: Checked ✅
- Edit: Unchecked ❌
- Delete: Unchecked ❌
- View All / Modify All: Unchecked ❌
- **Tab Settings:** *Default On*

Click **Save**. Rahul can now view and create employee records, but cannot edit or delete any records.

STEP 03 **Configure Amit's Access (Full CRUD Access)**

```
Setup ➔ Profiles (or Permission Sets) ➔ [Amit's Profile / Perm Set] ➔ Object Settings ➔ Employees
```

Click **Edit** and grant full CRUD capability:

- Read: Checked ✅
- Create: Checked ✅
- Edit: Checked ✅
- Delete: Checked ✅
- **Tab Settings:** *Default On*

Click **Save**. Amit now has complete CRUD authority over Employee records.

STEP 04 **Configure Priya's Access (Complete Lockout - No Access)**

```
Setup ➔ Profiles ➔ [Priya's Profile] ➔ Object Settings ➔ Employees
```

Click **Edit** and ensure all object checkboxes are disabled:

- Read: Unchecked ❌
- Create: Unchecked ❌
- Edit: Unchecked ❌
- Delete: Unchecked ❌
- **Tab Settings:** *Tab Hidden*

Click **Save**. Priya is completely locked out of the Employee object.

10. User Testing Verification Matrix

Now log in as each user (or test via appropriate admin view) to observe the exact behavioral outcome across the Salesforce user interface:

USER	ASSIGNED RIGHTS	VIEW TAB / RECS	"NEW" BUTTON	"EDIT" BUTTON	"DELETE" ACTION
Rahul (Sales Rep)	Read + Create	Visible ✅	Visible ✅	Hidden ❌	Hidden ❌
Amit (Operations)	Full CRUD	Visible ✅	Visible ✅	Visible ✅	Visible ✅
Priya (Finance)	No Access	Hidden ❌	Hidden ❌	Hidden ❌	Hidden ❌

> **Admin Testing Shortcut:**
> If your Developer Edition only has 2 standard test users available (Rahul and Priya), you can use your primary **System Administrator** account as the full CRUD/Modify All perspective without needing to purchase or activate extra licenses!

11. Key Takeaways & Episode Summary

1. CRUD is Fundamental

Create, Read, Edit, Delete determine whether a user can perform core database operations on an object. Read is required for Edit and Delete.

2. View All & Modify All Bypass Sharing

These object-level permissions grant total read or edit/delete visibility across all records of that specific object, ignoring OWD and sharing rules.

3. Object vs. System Scope

View All / Modify All apply to **one object**. *View All Data / Modify All Data* are system permissions that apply to **every object across the org**.

4. Object Access vs. Record Access

Object-Level Security answers *"What can you do with the object?"* while Record-Level Security answers *"Which individual records can you see?"*.

12. Next Episode Preview

📺 Coming Up in Episode 36: Field-Level Security (FLS)

Now that we know how to unlock or lock an entire object, we zoom into individual columns of data! In Episode 36, we explore **Field-Level Security (FLS)**. We'll answer: *"Even if Rahul has Read access to the Employee object, should he be able to see the **Salary** field?"* We'll master **Visible**, **Read-Only**, and page layout interactions.